

# Chinemerem C. Okezie

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## Professional Summary

PhD candidate in Civil and Environmental Engineering with over three years of interdisciplinary research experience in subsurface energy systems. Skilled in data analysis, analytical chemistry, geomechanics, fluid-rock interaction, reservoir engineering, and petrophysics. Research projects have focused on carbon capture and storage (CCS), hydrogen storage, serpentinization, geologic hydrogen, and geothermal energy. Currently seeking internship opportunities in roles related to subsurface energy systems.

## Education

PhD in Civil and Environmental Engineering Sep 2024 – Present

New Jersey Institute of Technology, Newark, NJ

Advisor: Dr. Oladoyin Kolawole

Relevant Coursework: Mechanics of Materials, Rock Mechanics, Deep Underground Engineering & Resources

MSc in Energy and Earth Resources Aug 2022 – Aug 2024

Jackson School of Geosciences, The University of Texas at Austin, Austin, TX

Thesis: Calibrating Performance Predictions for Large-Scale Injection  
[<https://doi.org/10.26153/tsw/57870>]

Advisors: Dr. Alex Bump, Dr. Susan Hovorka

Relevant Coursework: Data Analytics and Geostatistics, Machine Learning, Python for Geoscience Research

BEng in Petroleum Engineering Sep 2013 – May 2019

The Federal University of Technology, Owerri, Nigeria

Thesis: Numerical Simulation of Waterflooding: a case study of the Alpha Field in the Niger Delta

## Research Experience

Graduate Research Assistant, GGES Lab, New Jersey Institute of Technology Sep 2024 – Present

- Investigating the reaction pathways of the serpentinization reaction to maximize the potential of geologic hydrogen production.
- Utilizing analytical techniques such as scanning electron microscopy coupled with energy dispersive X-ray spectroscopy (SEM-EDS), X-ray Diffraction (XRD), Thermogravimetric Analysis (TGA), and Raman Spectroscopy, to assess the reaction progress and mineralogical evolution.
- Utilizing geochemical models to assess the reaction kinetics and pathways, to support the experimental results.
- Investigating the geomechanical implications of large-scale geologic hydrogen production via serpentinization.

- Investigated the concepts of sub-compartmentalization, hydraulic communication between wells, and formation damage in the onshore Texas Gulf Coast for large-scale CCS projects.
- Utilized Python, Petrel, ArcGIS, Petra, Harmony, and Excel to manage and analyze large saltwater disposal and well data to calibrate the current predictions for large-scale injection of CO<sub>2</sub> into geologic formations in the onshore Texas Gulf Coast
- Assessed and predicted the potential of carbon storage in offshore US Gulf Coast using pressure and production data from natural gas wells.

## Publications

### Conference Papers

- Okezie, C. C., Bump, A., & Hovorka, S. (2024). *Calibrating Performance Predictions for Large-Scale Injection* (SSRN Scholarly Paper No. 5065593). Social Science Research Network. <https://doi.org/10.2139/ssrn.5065593>
- Hovorka, S., Smye, K., Hennings, P.H., Bump, A.P., Okezie, C.C., & Luciano, A.K. (2024). *A tale of two geologic systems – comparing the impacts of injection into the Permian Basin to the Gulf of Mexico* (SSRN Scholarly Paper No. 5064850). Social Science Research Network. <https://doi.org/10.2139/ssrn.5064850>

### Manuscripts under Review

- Okezie, C.C. & Bump, A.P. *Calibrating Large-Scale Injection: Using Saltwater Disposal Experience to Predict CCS Performance on the Texas Gulf Coast*. Under review at the International Journal of Greenhouse Gas Control.
- Okezie, C.C. & Kolawole, O. *Geochemical and Geomechanical Risks of Subsurface Hydrogen Storage: Current Status and Future Opportunities*. Under review at the Journal of Deep Underground Science and Engineering.

## Presentations

- Calibrating Performance Prediction for Large-Scale Injection. *Presented at the 2024 University of Texas Institute for Geophysics Discussion Hour*. [<https://www.youtube.com/watch?v=2J8SG-Mx2rU>]
- Calibrating Performance Prediction for Large-Scale Injection. *Presented at the 2024 CCUS Conference*
- The Promise of the Lower Cretaceous for CO<sub>2</sub> Storage. *Presented at the 2023 Gulf Coast Carbon Center Sponsors' Meeting*
- Prediction of Carbon Storage Potential using Pressure-Production Data. *Presented at the 2023 Gulf Coast Carbon Center Sponsors' Meeting*

## Technical Skills

### **Programming and Data Analysis:**

Python (NumPy, Pandas, Seaborn, Scikit-learn, Matplotlib, SciPy, Cartopy, GeoPandas, Pyproj), Power BI

**Statistical Analysis:**

Hypothesis Testing, Monte-Carlo Simulations, Bootstrapping

**Analytical Techniques:**

Scanning Electron Microscopy-Energy Dispersive X-ray Spectroscopy (SEM-EDS), X-ray Diffraction (XRD), Thermogravimetric Analysis (TGA), Raman Spectroscopy

**Laboratory and Experimental Methods:**

Triaxial Compression Test, Uniaxial Compression Test, Indirect Tensile Test (Brazilian Test), Hydrothermal Batch Reactor Experiments, Hydraulic Fracturing Experiments.

**Geoscience & Visualization Tools:**

Petrel, ArcGIS, Petra, Harmony, Eclipse, Google Earth Pro.

**Machine Learning and Data Science:**

Supervised (Regression, Classification) and Unsupervised Learning (Clustering)

**Modeling & Simulation:**

Finite Element Analysis (ANSYS), Geochemical modeling (EQ3/6, PHREEQC).

**Documentation and Workflow:**

Git/GitHub, LaTeX, Microsoft Office Suite

**Selected Projects**

AAPG Imperial Barrel Award (IBA) Competition

Team member, University of Texas at Austin

Spring 2024

- Evaluated the CO<sub>2</sub> Storage potential in the Otway Basin, Australia, using 3D seismic volumes and well logs from multiple wells.
- Delivered a technical presentation to a panel of industry experts.

AES Energy Innovation Challenge

Team member, University of Texas at Austin

Fall 2023

- Explored the idea of geothermal energy production and lithium extraction from the Salton Sea in California.
- Delivered a technical presentation to a panel of industry experts.

**Professional Experience**

Executive Trainee Consultant, Verraki, Lagos, Nigeria.

Oct 2019 – July 2022

- Utilized Excel and Power BI for data handling & analysis, and report building.
- Managed and delivered process maturity assessments for clients in West Africa.
- Managed software implementation and knowledge transfer to clients across West Africa.

**Awards & Honors**

Fellow, Texas Entrepreneurship Exchange for Energy (2023-2024)

Graduate Fellowship, Energy and Earth Resources Program, University of Texas at Austin (2022-2024)

Service Award, Society of Petroleum Engineers (2022)

Winner, Nigeria Institute of Physics Quiz Competition (2013)

International Physics Olympiad (2012)

## **Professional Affiliations & Leadership**

### **Leadership & Service Roles**

- Student Advisory Council, Kay Bailey Hutchison Energy Center (2023-2024)
- Vice President of Professional Development & Editorial Council, Longhorn Energy Club (2022-2024)
- Volunteer, International Meeting for Applied Geoscience & Energy (2023)
- Communication & Outreach Chairperson, Society of Petroleum Engineers, Lagos (2021-2022)

### **Professional Membership**

- Member, American Association of Petroleum Geoscientists
- Member, Energy Resources Group
- Member, Edison Electric Institute
- Member, Solar Energy Industries Association
- Member, SDG 7 Youth Constituency